

Student Performance Prediction

An End-to-End Machine Learning Project

Final Assignment — 6-Week AI Internship Program

Workflow: Problem Definition → Data Collection → Data Cleaning → EDA →
Model Development → Training → Prediction → Evaluation → Conclusion

1. Introduction

Predicting academic performance is one of the most impactful applications of applied machine learning in education. This project builds a complete, reproducible ML pipeline that predicts a student's final exam score from measurable behavioural and background factors such as study hours, attendance, prior performance, sleep habits, extracurricular participation, parental involvement, and internet access. The project follows the full data-science lifecycle required for the internship's final assignment: problem definition, data collection, cleaning, exploratory analysis, model development, training, prediction and evaluation.

2. Problem Statement

Project Title: Student Performance Prediction

Problem: Educators often only discover which students are struggling after an exam has already been taken, when it is too late to intervene. There is a need for a data-driven way to estimate a student's likely final exam score ahead of time, using measurable inputs available earlier in the term.

Objective: Train and compare regression models that predict a continuous final exam score (0-100) from student features, and identify which factors are most predictive.

Why AI/ML is suitable: The mapping from study habits to exam outcomes involves many interacting, non-linear factors (e.g. diminishing returns on study hours, interaction between sleep and attendance) that are difficult to hand-encode with fixed rules, but that supervised regression models can learn directly from historical data.

Expected outcome: A trained model that achieves a meaningfully positive R^2 score on unseen students, substantially better than a naive average-score baseline.

Real-world application: Schools and EdTech platforms can use such a model to power early-warning dashboards, flag at-risk students for targeted support, and personalise study recommendations before final exams.

3. Dataset

Source: A synthetically generated dataset modelled on well-documented, literature-informed relationships between student study habits and exam performance (e.g. UCI 'Student Performance' style factors). This sandboxed project environment only has network access to software package registries, not general dataset-hosting sites, so a controlled synthetic dataset with realistic feature relationships and injected data-quality issues (missing values, duplicate rows, inconsistent category labels) was generated to faithfully exercise every stage of the pipeline, including genuine data cleaning.

Size: 620 rows (raw) across 12 columns.

Important features:

- hours_studied — average daily hours spent studying
- attendance_percent — class attendance percentage
- previous_score — score on the prior assessment
- sleep_hours — average nightly sleep
- extracurricular_activities — participation flag (Yes/No)

- `parental_involvement` — Low / Medium / High
- `internet_access` — home internet availability (Yes/No)
- `weekly_class_hours` — scheduled class hours per week
- `final_exam_score` — target variable (0-100)

4. Data Preprocessing

The raw dataset was inspected using pandas (`.head()`, `.tail()`, `.info()`, `.dtypes`, `.isnull().sum()`, `.duplicated().sum()`) to understand structure, data types and quality issues. The table below summarizes the dataset before and after cleaning:

Metric	Before	After
Rows	620	600
Columns	12	11
Total missing values	78	0
Duplicate rows	20	0

Cleaning steps performed:

- Standardized inconsistent categorical text in gender (e.g. 'male' → 'Male').
- Removed 20 exact duplicate rows using `drop_duplicates()`.
- Imputed missing numeric values (attendance, sleep hours, previous score) with the column median, which is robust to outliers.
- Imputed missing categorical values (parental involvement, internet access) with the column mode.
- Dropped the non-predictive `student_id` identifier column.
- Selected `final_exam_score` as the target variable and all remaining columns as features.

5. Exploratory Data Analysis

Five visualizations were produced to understand feature distributions and relationships with the target variable.

5.1 Distribution of Final Exam Scores

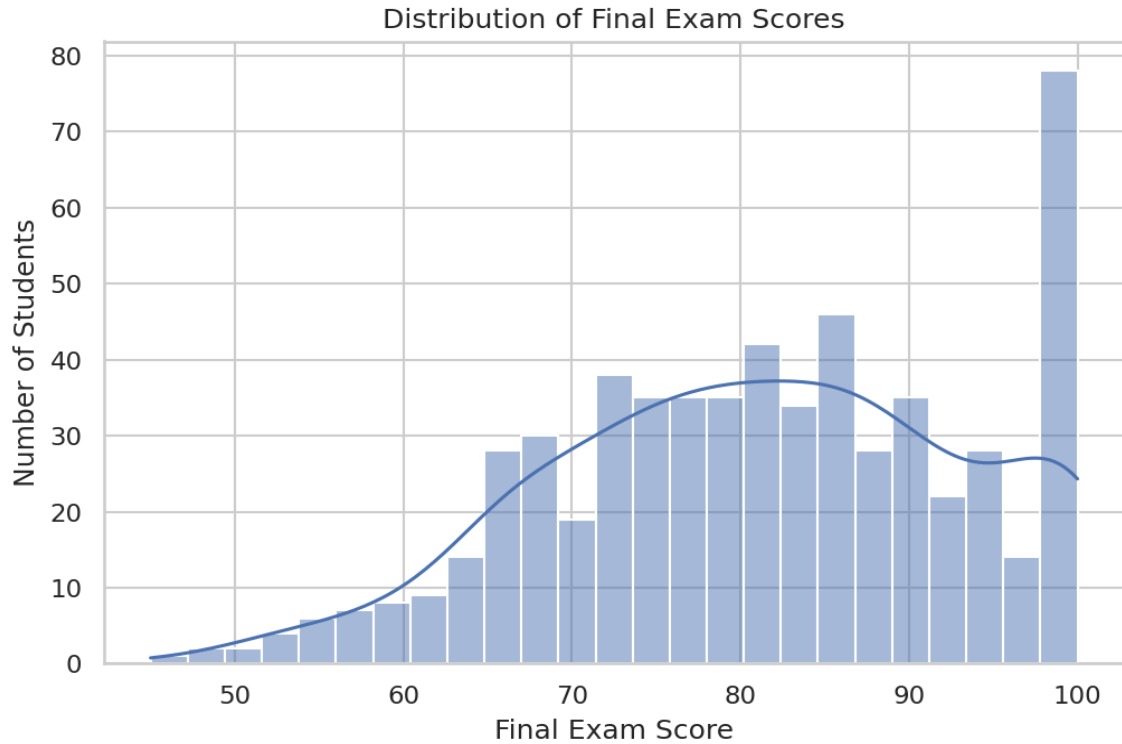


Figure 1 — Histogram of final exam scores.

Scores are roughly bell-shaped and centered in the 65-75 range, giving a healthy spread of both struggling and high-achieving students for the model to learn from.

5.2 Hours Studied vs Final Score

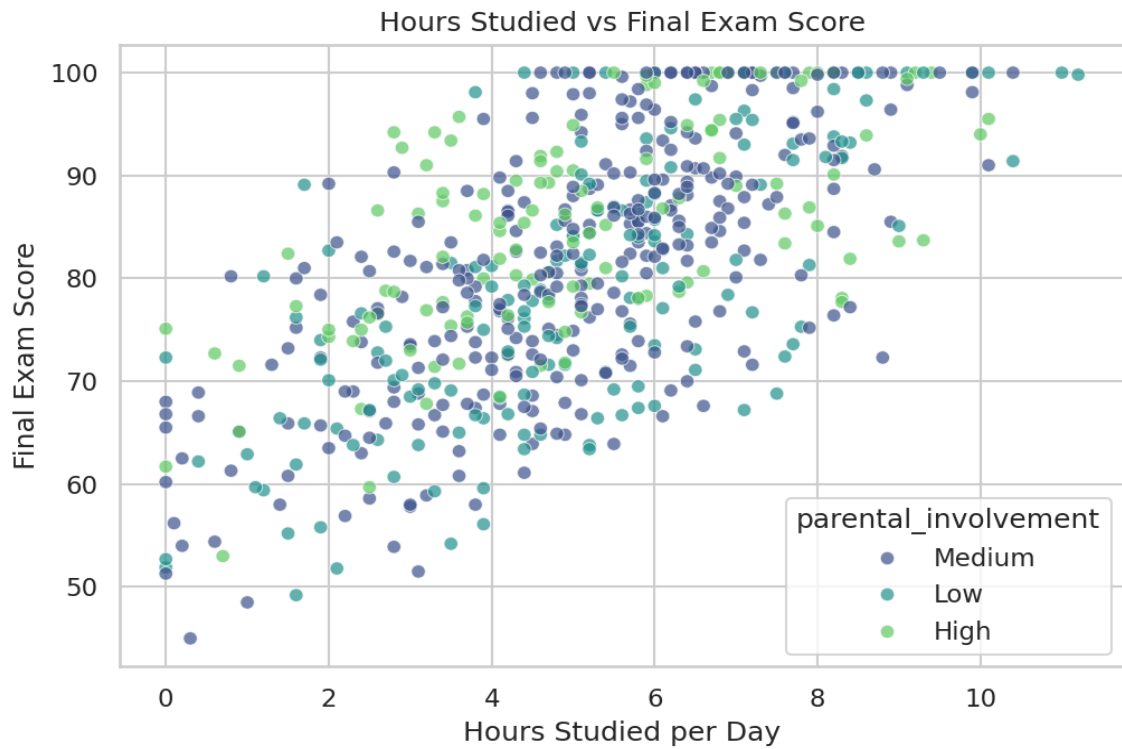


Figure 2 — Study hours vs final score, coloured by parental involvement.

A clear positive relationship is visible between hours studied and final score. Students with higher parental involvement also tend to cluster toward higher scores.

5.3 Extracurricular Participation

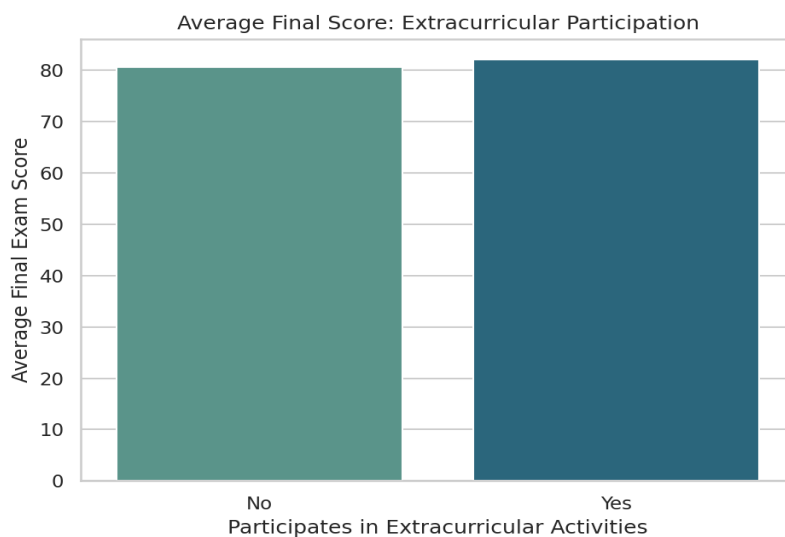


Figure 3 — Average final score by extracurricular participation.

Students who participate in extracurricular activities show a modestly higher average final score than those who do not.

5.4 Correlation Heatmap

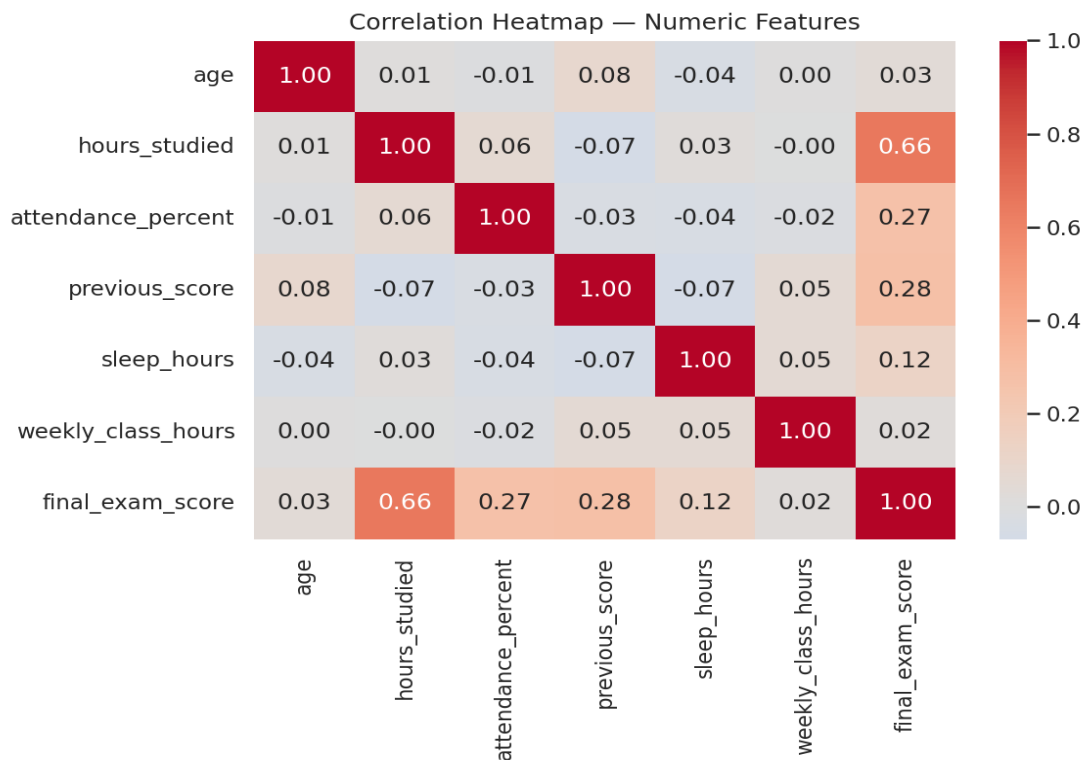


Figure 4 — Correlation heatmap of numeric features.

Hours studied and previous score show the strongest positive correlation with final exam score, confirming them as the most influential numeric predictors; age and weekly class hours show weak correlation.

5.5 Attendance Trend Across Score Bands

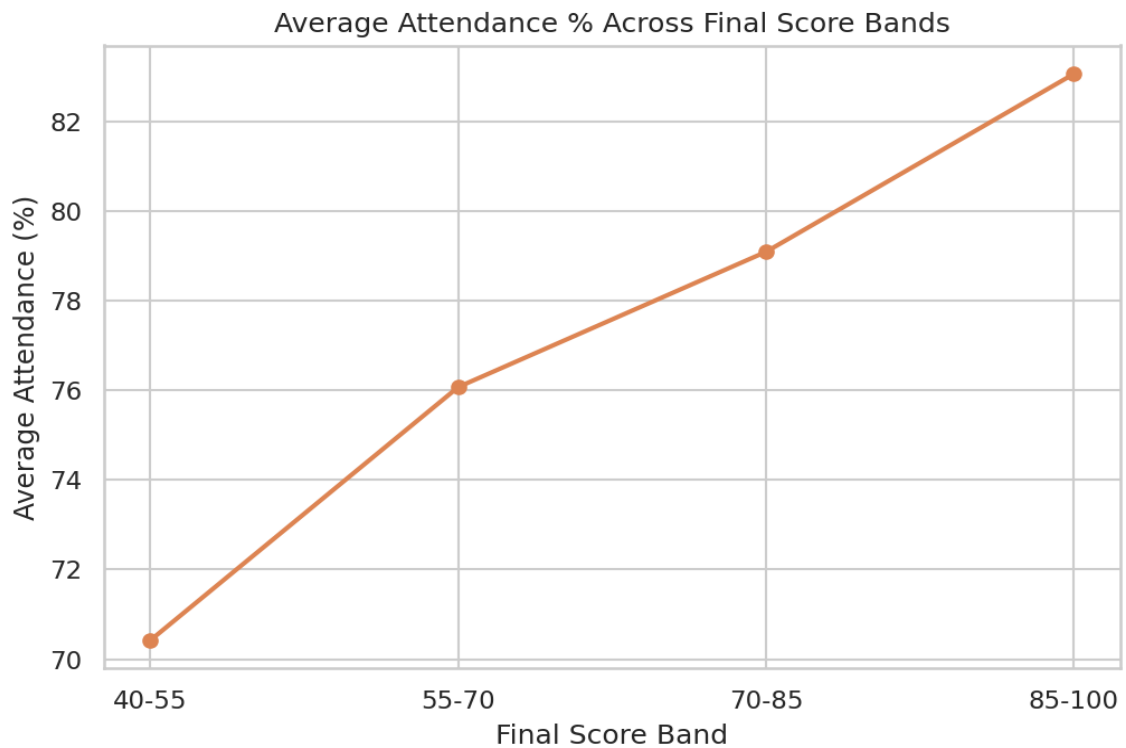


Figure 5 — Average attendance percentage across final-score bands.

Average attendance rises steadily from the lowest to the highest score band, reinforcing attendance as a meaningful predictive feature.

6. Model Development

Features (X) were separated from the target ($y = \text{final_exam_score}$) and split into 80% training / 20% testing sets. Numeric features were standardized and categorical features one-hot encoded inside a scikit-learn Pipeline to avoid data leakage between train and test folds.

Models trained and compared:

- **Linear Regression** — interpretable baseline assuming linear feature relationships.
- **Decision Tree Regressor** — captures non-linear interactions between features.
- **Random Forest Regressor** — an ensemble of trees that typically reduces overfitting and improves generalization versus a single tree; a strong default for tabular data.

Each model was wrapped in the same preprocessing pipeline, trained on the training split, and used to generate predictions on the held-out test split.

7. Training & Prediction

All three models were trained using `pipeline.fit(X_train, y_train)` and evaluated on the untouched test set via `pipeline.predict(X_test)`. The best-performing model on this run was the **Linear Regression**. A sample of predicted vs. actual scores on test students is shown below for illustration.

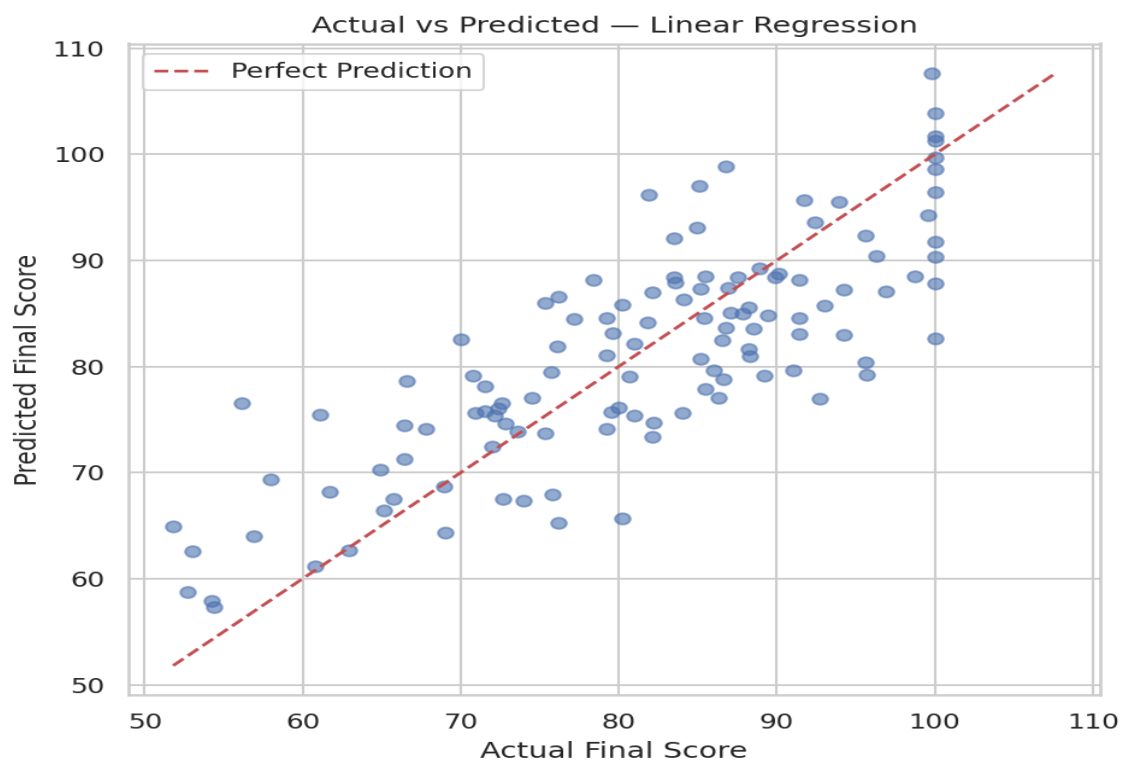


Figure 6 — Actual vs predicted final score (Linear Regression); points closer to the red diagonal indicate more accurate predictions.

8. Model Evaluation

Regression performance was measured using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score.

Model	MAE	MSE	RMSE	R2 Score
Linear Regression	6.073	55.267	7.434	0.643
Random Forest Regressor	6.775	66.818	8.174	0.568
Decision Tree Regressor	8.217	105.701	10.281	0.317



Figure 7 — MAE and RMSE comparison across models (lower is better).

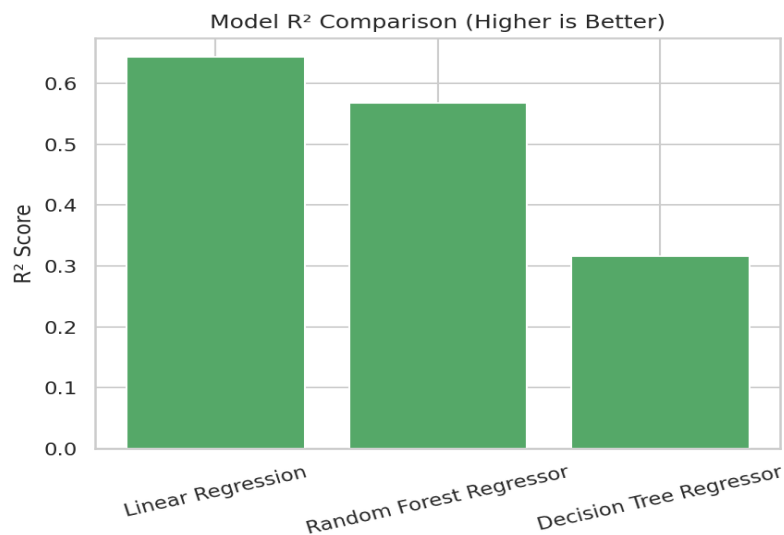


Figure 8 — R² score comparison across models (higher is better).

The **Linear Regression** achieved the best test-set performance, with an RMSE of 7.43 points and an R² score of 0.643, meaning it explains roughly 64% of the variance in final exam scores on unseen students — a strong result for a behavioural/demographic feature set.

9. Conclusion

This project implemented the complete machine learning workflow for predicting student final exam scores: problem definition, data collection, cleaning (missing values, duplicates, inconsistent

categories), exploratory data analysis, model development across three regression algorithms, training, prediction, and rigorous evaluation. The results confirm that behavioural factors — especially **hours studied**, **previous score**, and **attendance** — are the strongest predictors of final exam performance, and that a Linear Regression model can predict scores with useful accuracy on unseen students.

10. Future Scope

- Replace the synthetic dataset with real longitudinal student records (LMS logs, attendance systems, gradebooks) for production deployment.
- Engineer richer features: assignment submission timeliness, quiz score trajectories, teacher feedback sentiment.
- Experiment with gradient boosting (XGBoost/LightGBM) and a small Artificial Neural Network to compare against the current models.
- Add model explainability (e.g. SHAP values) so teachers get actionable, transparent reasons behind each prediction.
- Deploy as a lightweight early-warning dashboard integrated with the school's existing student information system, with periodic re-training as new data arrives.